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Research Topic: Mathematics – A trajectory that leads to a singularity could leave an observable geometric and topological footprint before reaching the critical regime.

Research Title: The Trace Hypothesis Toward Singularity

1.- Introduction and Historical Context

1.1 The Millennium Problem

What happens when an equation that describes the motion of a fluid stops describing it? The Navier-Stokes equations describe something as everyday as running water or moving air. And yet, in the three-dimensional incompressible case, a basic question remains open: starting from smooth initial data, can the solution develop a singularity in finite time? It is one of the seven Millennium Prize Problems of the Clay Mathematics Institute [1], [2], [3].

The hard part is not only that the velocity grows, but the struggle between two forces: nonlinearity, which concentrates vorticity, and viscosity, which smooths it out.

1.2 A New Starting Point

On September 8, 2026, OpenAI presented an analytical construction and a Lean Formalization in which a singularity forms in finite time from an initially smooth state, with an external force and finite energy throughout the entire evolution [4]. The mechanism described is almost visual: a vortex structure that spirals inward while it stretches and concentrates.

This result leaves an open question: does the observed geometry belong to the particular mechanism of that construction, or does it reveal a more general pattern on the way to a singularity?

1.3 The Emerging Question

This research shifts the focus from the critical instant to the pre-critical trajectory: instead of studying only $t \to T$, it will analyze the evolution of vorticity during $t<T$.

The central question is: is there a measurable geometric signature in the pre-critical evolution that can distinguish trajectories that lead to blow-up from those that remain regular?

1.4 Research Gap

Research on Navier-Stokes regularity has advanced along three partly independent paths.

1) The classical analytical criteria (the Beale-Kato-Majda criterion, formulated for the 3D Euler equations, and the continuation criteria for Navier-Stokes, which are related but different) give necessary conditions on the integrated vorticity, but say nothing about the geometric shape of the path toward that loss of regularity [5], [6].

2) The geometric constraints of Constantin, Fefferman, and Majda on the direction of vorticity can rule out singularity formation, but only as a sufficient condition, not as a signature that distinguishes pre-critical trajectories [7].

3) Topological data analysis has already been used to characterize the dynamics of vorticity fields in two-dimensional flows and convection, but has not yet been applied to anticipate a blow-up [8].

These three lines of work coexist without meeting in a common framework that answers whether there is a set of geometric and topological descriptors able to robustly distinguish a pre-critical trajectory from one that will remain regular. This gap, the lack of a pre-critical signature validated beyond a single case, is what this proposal aims to fill. The contribution is not a new blow-up construction, but the combination of geometric and topological descriptors, extending to three-dimensional flows a type of topological analysis that the cited works apply in 2D, to evaluate whether a reproducible and robust pre-critical signature exists.

2.- Hypothesis, Objectives, and Variables

2.1 From the Critical Point to the Trajectory

Vorticity dynamics in 3D contain a mechanism that does not exist in the plane: vortex stretching. For an incompressible fluid,

$$\partial\omega/\partial t + (u\cdot\nabla)\omega = (\omega\cdot\nabla)u + \nu\Delta\omega.$$

The term (ω·∇)u amplifies vorticity, while viscosity spreads it out and damps it; this balance is central to regularity theory [3], [9]. The geometry of vorticity also matters, as Constantin, Fefferman, and Majda point out [7].

2.2 The Geometric Signature Hypothesis

We propose that singular Navier-Stokes trajectories follow, before the critical time, a characteristic geometric evolution that can be expressed through the relationship between length L(t), effective radius R(t), vorticity intensity ωmax(t), and local strain rate.

The research will not assume that a single ratio is enough to predict blow-up: its falsifiable prediction is that the singular reference will be distinguishable from the controls using descriptive statistics and distance metrics, and will remain so under controlled perturbations.

$$\Lambda(t) = L(t) / R(t).$$

Λ(t) will be used as an initial descriptor of the relative elongation of the structure. Its growth is not assumed as a law: it may diverge, stay bounded, or fail to separate the reference from the controls; only a systematic and reproducible separation would support the hypothesis.

2.3 Research Objectives

General objective: to determine whether the pre-critical evolution contains a geometric and topological signature that distinguishes the singular reference trajectory from regular ones under controlled perturbations.

Specific objectives: (1) to describe the time evolution of L(t), R(t), ωmax(t), and the strain-rate tensor S in the singular reference trajectory and in the controls; (2) to compare its geometric dynamics with the controls using descriptive statistics and distance metrics, looking for windows of systematic and reproducible separation; and (3) to evaluate its robustness through controlled perturbations and topological descriptors based on persistent homology.

2.4 Observable Variables

L(t): length of the centerline of the vortex structure. R(t): effective radius of its cross-sectional vorticity distribution. ωmax(t): maximum norm of the vorticity.

The local strain rate will be recorded to link geometry with the stretching mechanism; the sensitivity of R(t) to the threshold is discussed in Section 3.2.

3.- Computational Design and Methodology

The methodology separates a required core (L(t), R(t), ωmax(t), S, and the controls) from an optional add-on (TDA), which is used only if geometry alone is not enough: vorticity → structure → L(t), R(t), S → (optional TDA) → comparison with controls.

3.1 Field Preparation

The experiments start from the field u(x,t) described in Sections 3.3 and 3.4, with vorticity ω = ∇×u and strain-rate tensor $S = 1/2(\nabla u + \nabla u^T)$, which form the basis of the geometric descriptors.

The reproduction will use a cubic periodic domain with a pseudo-spectral method and dealiasing, Leray projection for incompressibility, and third- to fourth-order Runge-Kutta time integration with an adaptive time step (CFL condition); ν and f will be set according to [4], and the reproduction will be validated through energy, enstrophy, and the growth of ωmax(t) compared with what is reported in [4], following the numerical diagnostic criteria of [10].

3.2 Extraction of Vortex Structures

The dominant vortex structure at each instant is defined using a relative threshold on vorticity:

$$|\omega(x,t)| > \alpha \cdot \omega max(t), \quad \alpha \in (0,1).$$

L(t) and R(t) are computed over that region. Since α is arbitrary, its effect on L(t), R(t), and Λ(t) will be explored, so that the results do not depend on a single threshold.

3.3 Control Group

The main control (Singular 3D versus regular 3D) will be a family of 3D simulations with the same domain, resolution, and viscosity as the reference, smooth initial conditions, and controlled variations in amplitude and forcing; any trajectory whose energy and enstrophy stay bounded and whose ωmax does not diverge will be classified as regular. This family will provide the baseline for L, R, ωmax, and the derived quantities [1], [11]. Two-dimensional configurations, which lack 3D vortex stretching, will serve only as methodological validation, not as the main group.

3.4 Singular Reference Trajectory

A numerical reproduction (not a direct implementation) of OpenAI's analytical construction [4] will be carried out, saving velocity and vorticity at each instant. This reproduction is a single singular reference trajectory: Section 4.4 will evaluate it against the controls and the perturbations of Section 3.5 using descriptive statistics and distance metrics, never as a second symmetric group. From the vortex structure, the following time series will be built:

$$\{\Lambda(t), R(t), L(t), \omega max(t)\}.$$

The analysis will focus on the pre-critical regime, taking care not to mistake numerical divergence caused by insufficient resolution near the critical time for physical evidence; the goal is to find the window where the dynamics move away from the controls.

3.5 Structural Stability Test

To find out whether the signature belongs only to this particular construction or reflects a more robust mechanism, controlled perturbations will be introduced:

Initial field: perturbations $u_0 = \varepsilon v(x)$, with $\nabla \cdot v = 0$ and small ε, since the construction starts from rest rather than from a geometry that can be perturbed.

Viscosity: systematic variation of $\nu > 0$ within a defined range.

Forcing: perturbations $f \rightarrow f + \varepsilon g$ that are smooth and bounded, with no external singularity.

For each perturbation, the normalized Λ(t), R(t), and ωmax(t) will be compared: robustness will be supported if the signature is preserved; if it disappears, the signature depends on the reference configuration.

4.- Mathematical Formalization and Topological Analysis

4.1 Strain-Rate Tensor

The proposed geometry is linked to local dynamics through the symmetric strain-rate tensor

$$S = 1/2 (\nabla u + \nabla u^T).$$

Its spectral decomposition shows the directions of compression and stretching; this study will examine whether the largest eigenvalue of S and the growth of Λ(t) show a reproducible relationship before the critical regime [9]. For reference, in three-dimensional turbulence, vorticity tends to align with the intermediate eigenvector of S [12].

4.2 Connection with Beale-Kato-Majda

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The Beale-Kato-Majda criterion states that if an initially smooth solution loses regularity in 3D Euler, the maximum vorticity becomes non-integrable in time [5]; for Navier-Stokes, there are continuation criteria based on vorticity [6].

The research will not assume that $\Lambda(t) \to \infty$ is equivalent to a BKM criterion; instead, it will study whether an observable geometric evolution comes before the relevant growth in vorticity:

$$\Lambda(t), \quad \omega max(t), \quad \int_0^t \omega max(\tau) d\tau.$$

The goal is to evaluate geometry as a pre-critical descriptor, not as an analytical criterion.

4.3 Topological Data Analysis

Length and radius can lose information when vortex structures change their connectivity; TDA will be used only in that case, as a conditional add-on and not as a default technique. Topological data analysis (TDA), specifically persistent homology, will be used [13], [14]: studies of vortex-dominated flows show that this technique identifies vortex structures and that bottleneck and Wasserstein distances measure topological changes [8], [15].

Filtrations will be built from the vorticity field, along with their persistence diagrams, to evaluate whether the singular reference trajectory shows a topological evolution that is distinguishable from the controls and whether it improves the classification based on L/R.

The procedure follows a clear chain: a filtration is built from the vorticity field; from it, the persistent homology and its diagram are obtained; and from the diagram, the topological component of the vector X_r in Section 4.4 is extracted.

4.4 Metrics and Statistical Analysis

The comparison will not rely on visual inspection of the time series, but on a summary vector for each trajectory, $X_r = [\langle\Lambda\rangle, \max \Lambda, d\Lambda/dt, \langle\omega max\rangle, \max \omega max, \langle\lambda max(S)\rangle, \text{TDA}]$, so that the statistical unit is the trajectory and not each instant, which avoids pseudoreplication. For the controls, the mean and standard deviation of each component will be calculated, and the reference X_r will be placed within that distribution (empirical percentile, Mahalanobis distance); the same vector, computed for each perturbation in Section 3.5, will allow a non-parametric test (Mann-Whitney) if there are enough cases, and will include the Wasserstein distance as its topological component [8]. Formal inference (Mann-Whitney, percentiles) is applied to the families of controls and perturbations; the singular reference is treated as a single case compared against those distributions, not as a sample from a population.

The sensitivity of these metrics to the threshold α and to the perturbation amplitude will be reported, to show that the separation is systematic and not a parameter artifact. A signature will be considered discriminative if the reference X_r exceeds the 95th percentile of the controls in at least two components, remains stable under ±20% changes in α, and persists under the perturbations of Section 3.5.

5.- Scientific Impact and Conclusions

5.1 Expected Outcome

We expect to determine whether there is a region of the feature space $F = (\Lambda, \omega max, \lambda max(S),$ TDA vector) where the singular reference trajectory systematically separates from the control set during the pre-critical regime. A positive result would be a reproducible separation that is robust to the perturbations of Section 3.5; a negative result would show that this separation does not exist or depends strongly on the configuration studied. In both cases, the result will be reported in terms of the metrics of Section 4.4, not as a qualitative claim.

5.2 Limitations

This proposal openly acknowledges its methodological limitations: near the critical time T, the available resolution may not be enough to distinguish physical growth from a numerical artifact, so the analysis will use a pre-critical window far from that limit.

The extraction of L(t) and R(t) depends on the threshold α (Section 3.2), and this dependence cannot be fully removed. The robustness of the signature will be evaluated only within the family of perturbations of Section 3.5, so that conclusion will be limited to it. Finally, a single blow-up mechanism, even if well documented, does not allow claims of universality: the results are evidence about that construction and its neighborhood, not a general theorem.

5.3 Feasibility

The project is organized into five phases: numerical reproduction of the OpenAI construction [4]; extraction of vortex structures and computation of geometric descriptors; construction of the controls and initial statistical comparison; controlled perturbations and robustness evaluation; and addition of topological data analysis and joint evaluation of F. Partial results can already be interpreted from the third phase, without depending on the later ones.

5.4 If the Hypothesis Is Correct

If a pre-critical geometric signature that withstands controlled perturbations is identified, the path to a singularity would have a more robust structure than that of an isolated case: it would point to a principle for classifying trajectories using pre-critical geometric and topological variables.

5.5 If the Hypothesis Is False

A negative result would also teach something: if the signature fades under small perturbations, the geometry depended on the configuration studied. The proposal therefore keeps a falsifiable logic: whether the signature persists or fades, new knowledge is gained.

5.6 Conclusion

This research does not aim to replace the analytical regularity criteria, but to explore a complementary dimension: the geometry of the path toward the critical regime. The OpenAI construction provides a singular case; classical theory, its analytical criteria; and TDA, a way to turn complex flow structures into measurable objects.

The question that connects these three levels is:

Can the path to a singularity be recognized geometrically before it is reached?

Answering it would show whether the blow-up studied is a specific configuration or reveals a more general structure in three-dimensional fluid dynamics. And if a pre-critical trajectory could be characterized robustly, the question would change in nature: it would no longer be only whether a singularity occurs, but how the system, step by step, evolves toward it.

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